Dynamic comparison of Monod and substrate inhibition models

Recombinant lipase production
and oxygen transfer design

GROUP 21

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23B0353

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23B0345

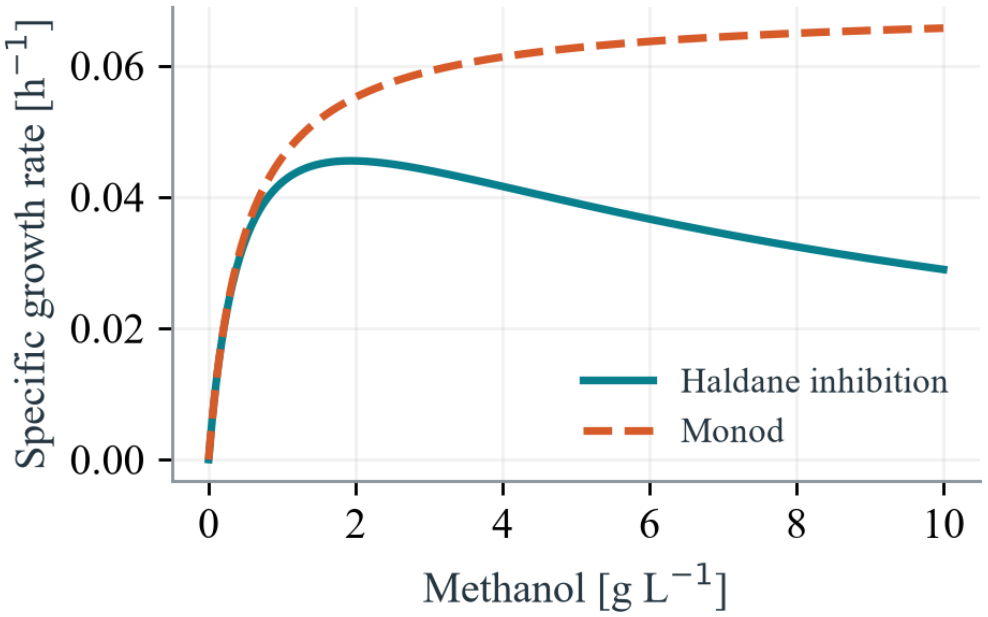
Diggaj Maheshwari

23B0358

Methanol is both feed and inducer

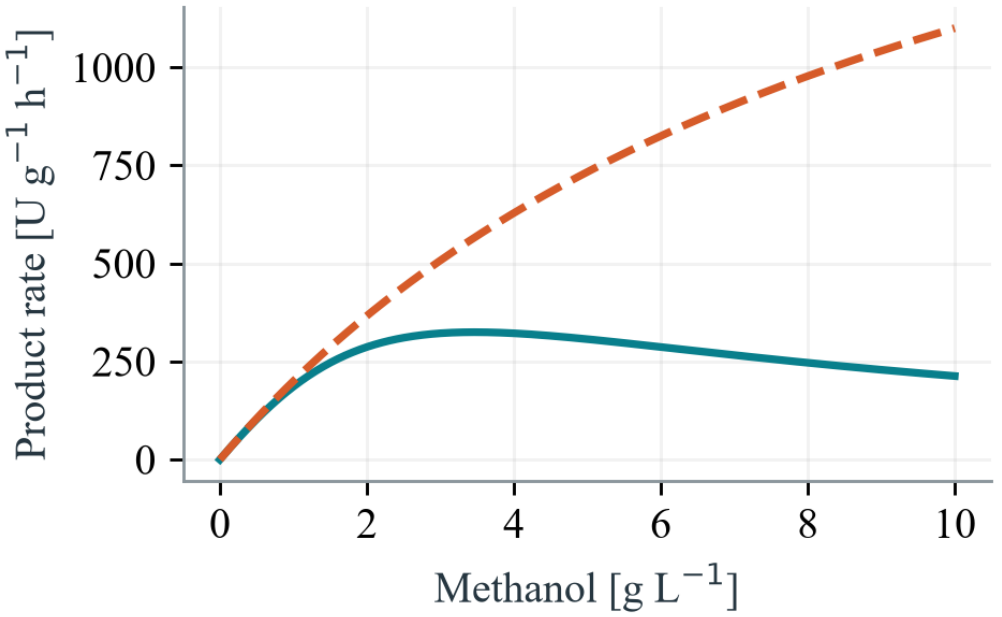
Pichia pastoris

Methanol induced yeast



Rhizopus oryzae lipase

Extracellular industrial enzyme



5 L vessel. 30 °C. pH 5.5. DO 25% air.

Four balances define the process

V, B, M, A

Volume, biomass, methanol, activity

$$\frac{dV}{dt} = \alpha F$$

$$\frac{dB}{dt} = \mu B$$

$$\frac{dM}{dt} = F - q_s B$$

$$\frac{dA}{dt} = q_p B$$

$\alpha = 1/792 + 0.12/200$

Includes nutrient feed volume

CF **Constant feed**

14 g/h

CM **Prescribed growth target**

$\mu = 0.03\ h^{-1}$

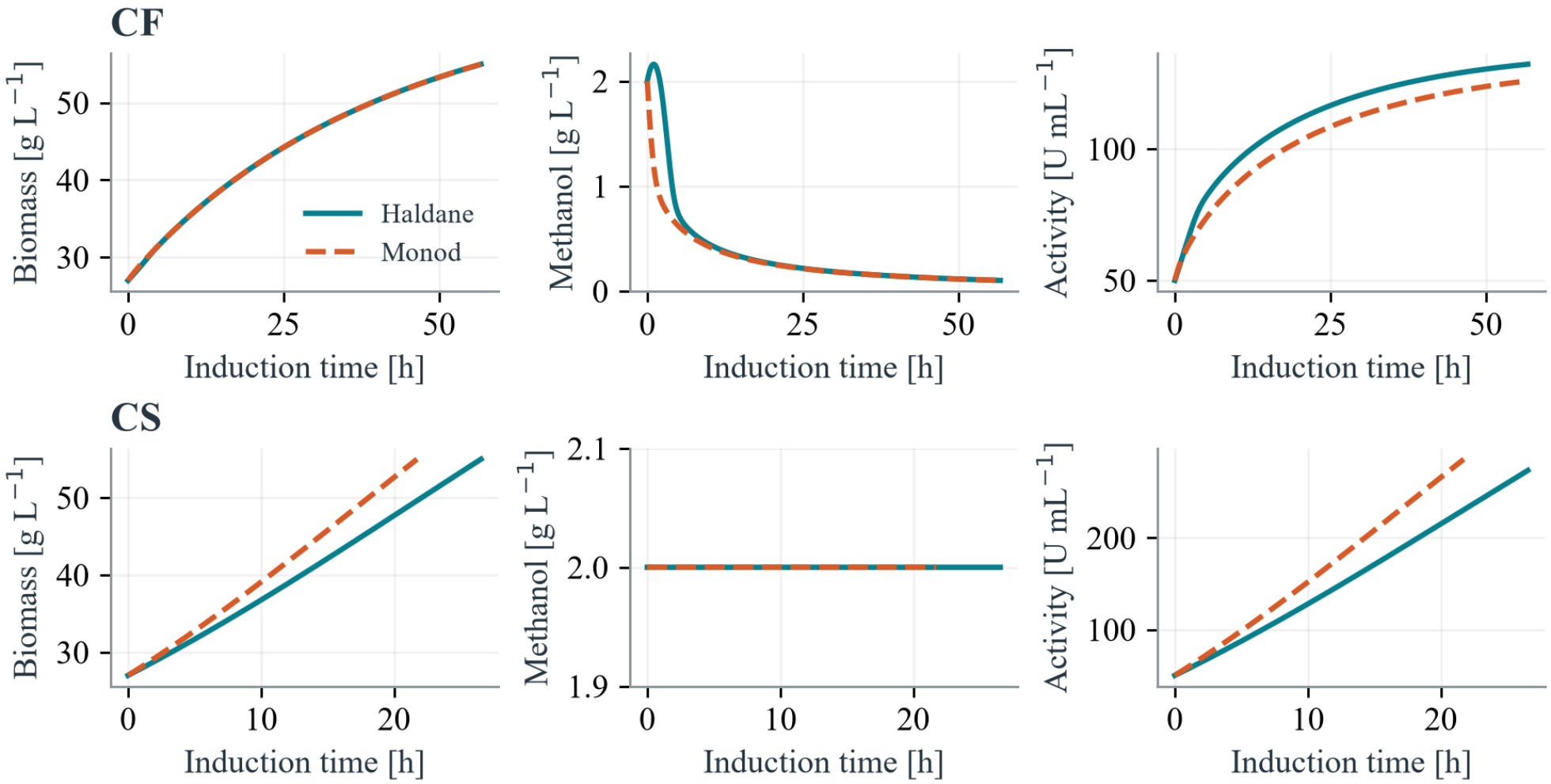
LM **Ramped growth target**

0.01 to 0.04 h^{-1} over 60 h

CS **Constant methanol**

Reference S = 2 g/L

CS more than doubles CF activity



Setpoints change the operating decision

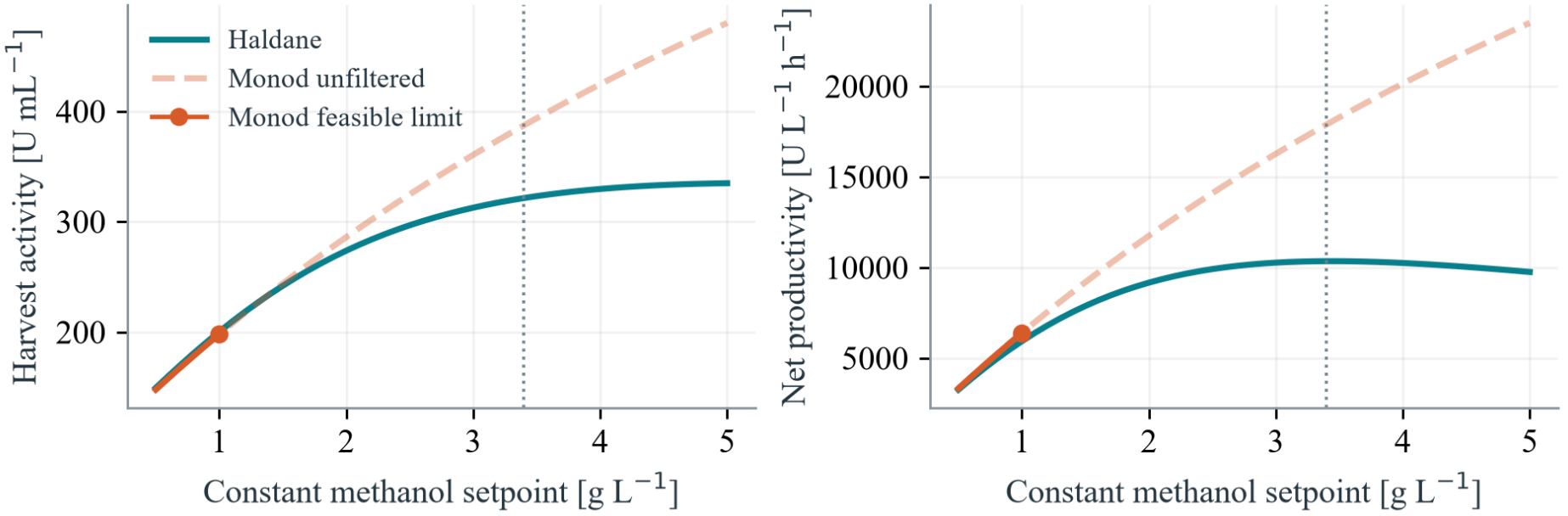
3.4

g/L candidate setpoint

Predicted activity and productivity

321.35 U/mL

10,349 U/L/h



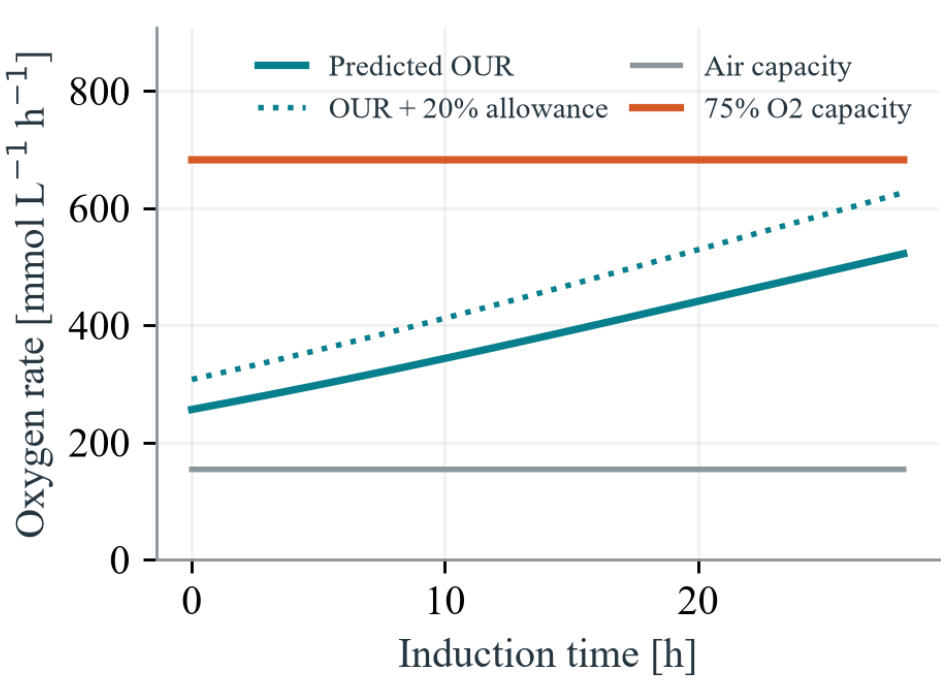
Oxygen supply sets an operating limit

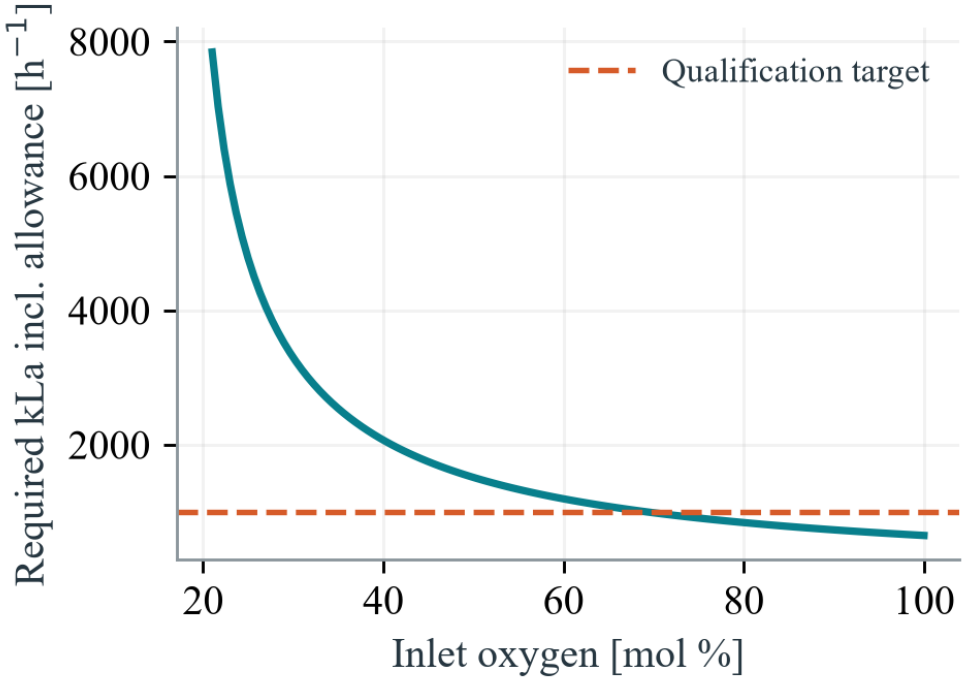
909

required kLa, h^{-1}

1,000

broth qualification target, h^{-1}





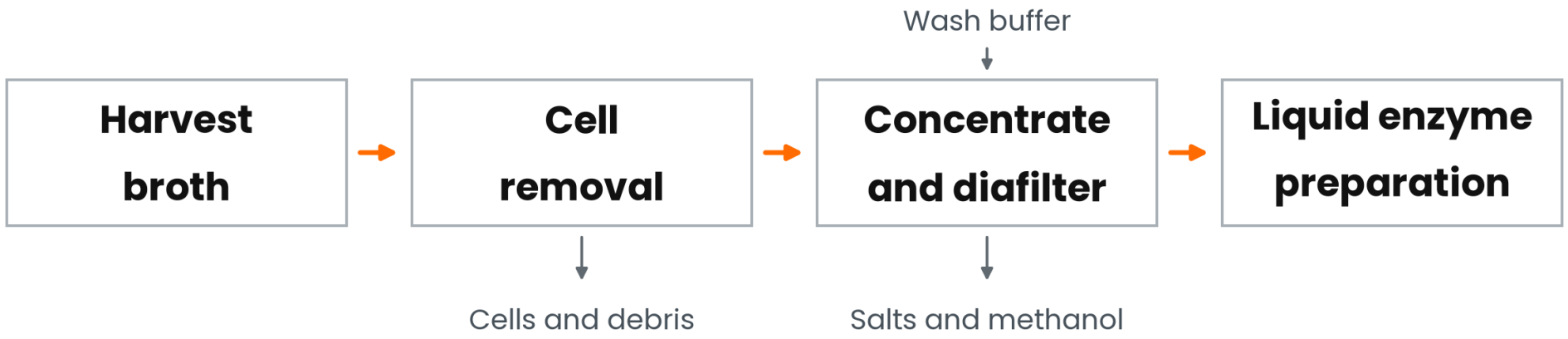
Recovery determines the activity cost

₹10,016

illustrative batch budget

70%

assumed activity recovery



₹13,487 per million recovered U

Industrial transfer needs pilot oxygen and mixing checks.